The Rational Clinical Examination

# Does This Patient Have an Exudative Pleural Effusion?
## The Rational Clinical Examination Systematic Review

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**IMPORTANCE** Thoracentesis is performed to identify the cause of a pleural effusion. Although generally safe, thoracentesis may be complicated by transient hypoxemia, bleeding, patient discomfort, reexpansion pulmonary edema, and pneumothorax.

**OBJECTIVE** To identify the best means for differentiating between transudative and exudative effusions and also to identify thoracentesis techniques for minimizing the risk of complications by performing a systematic review the evidence.

**DATA SOURCES** We searched The Cochrane Library, MEDLINE, and Embase from inception to February 2014 to identify relevant studies.

**STUDY SELECTION** We included randomized and observational studies of adult patients undergoing thoracentesis that examined diagnostic tests for differentiating exudates from transudates and evaluated thoracentesis techniques associated with a successful procedure with minimal complications.

**DATA EXTRACTION AND SYNTHESIS** Two investigators independently appraised study quality and extracted data from studies of laboratory diagnosis of pleural effusion for calculation of likelihood ratios (LRs; n = 48 studies) and factors affecting adverse event rates (n = 37 studies).

**RESULTS** The diagnosis of an exudate was most accurate if cholesterol in the pleural fluid was greater than 55 mg/dL (LR range, 7.1-250), lactate dehydrogenase (LDH) was greater than 200 U/L (LR, 18; 95% CI, 6.8-46), or the ratio of pleural fluid cholesterol to serum cholesterol was greater than 0.3 (LR, 14; 95% CI, 5.5-38). A diagnosis of exudate was less likely when all Light's criteria (a ratio of pleural fluid protein to serum protein >0.5, a ratio of pleural fluid LDH to serum LDH >0.6, or pleural fluid LDH >two-thirds the upper limit of normal for serum LDH) were absent (LR, 0.04; 95% CI, 0.02-0.11). The most common complication of thoracentesis was pneumothorax, which occurred in 6.0% of cases (95% CI, 4.0%-7.0%). Chest tube placement was required in 2.0% of procedures (95% CI, 0.99%-2.9%) in which a patient was determined to have radiographic evidence of a pneumothorax. With ultrasound, a radiologist's marking the needle insertion site was not associated with decreased pneumothorax events (skin marking vs no skin marking odds ratio [OR], 0.37; 95% CI, 0.08-1.7). Use of ultrasound by any experienced practitioner also was not associated with decreased pneumothorax events (OR, 0.55; 95% CI, 0.06-5.3).

**CONCLUSIONS AND RELEVANCE** Light's criteria, cholesterol and pleural fluid LDH levels, and the pleural fluid cholesterol-to-serum ratio are the most accurate diagnostic indicators for pleural exudates. Ultrasound skin marking by a radiologist or ultrasound-guided thoracentesis were not associated with a decrease in pneumothorax events.

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## Clinical Scenario

A 74-year-old man is admitted to the hospital for cough, dyspnea, and fever. He has a history of congestive heart failure, a temperature of 39°C, respiratory rate of 26/min, oxygen saturation of 86% on ambient air, decreased breath sounds, dullness to percussion in the left lower thorax, and a white blood cell count of 19 600/μL. Chest radiography shows opacification of the left hemithorax consistent with a pleural effusion. A thoracentesis is planned to distinguish between an exudative and transudative effusion because the causes and management of pleural effusions differ when they are transudates vs exudates; for example, the effusion in this patient could be from heart failure or a parapneumonic process. The appropriate tests of the fluid must be ordered to establish the diagnosis and also to determine the techniques that best ensure procedural success with minimal complications.

## Importance of a Diagnostic Thoracentesis

Pleural effusions are an accumulation of fluid in the pleural space from pulmonary, pleural, or extrapulmonary diseases. Information from the patient's medical history and physical examination can help differentiate between transudative and exudative effusions, but the distinction between the 2 types can only be established by direct examination of the pleural fluid.[1] Light's criteria are often used to establish a diagnosis of exudative effusion.[2] An exudate is present when at least 1 of the following is observed: a ratio of pleural fluid protein to serum protein greater than 0.5; a ratio of pleural fluid lactate dehydrogenase (LDH) to serum LDH greater than 0.6; or pleural fluid LDH greater than two-thirds the upper limit of normal for serum LDH (previously, a cutoff value of 200 U/L was used).[2] When Light's criteria are used,[3-7] 7.8% to 15% of effusions are incorrectly classified as exudates when they are really transudates, as defined by a microbiological or pathological diagnosis (eg, malignant cells or specific organism in pleural fluid) or clinical response to treatment (eg, resolution with diuresis).

### Adverse Events Associated With Thoracentesis

Thoracentesis is an invasive procedure that is associated with complications including puncture site pain, pneumothorax, bleeding (eg, hematoma, laceration of intercostal artery, and liver or spleen puncture), transient hypoxemia, reexpansion pulmonary edema, vasovagal events, malignant seeding of the needle tract, and adverse reactions related to anesthetic or topical antiseptic solutions used during the procedure.[8-10] Although uncommon, intrapleural retention of catheter fragments can also occur.[11] Pneumothorax is the most important complication and occurs in approximately 1.3% to 26% of procedures.[12] Observational studies suggest that the technique used for thoracentesis influences the types and frequency of subsequent complications. It is generally believed that ultrasound guidance decreases the risk of pneumothorax.[13,14]

Little is known about the safety of thoracentesis in patients with coagulation abnormalities such as prolonged prothrombin time, prolonged partial thromboplastin time, or thrombocytopenia. One case series showed no increase in bleeding complications associated with thoracentesis among patients with mild elevation of prolonged prothrombin time or prolonged partial thromboplastin time (up to twice the midpoint normal range) or with platelet counts between 50 × $10^3$/μL and 99 × $10^3$/μL.[15] In a retrospective cohort study (n = 1076), the risk for bleeding was not increased for patients with international normalized ratio greater than 3 (n = 32) or platelet count less than 25 × $10^3$/μL (n = 12) when radiologists using ultrasound guidance performed all of the procedures.[16] Patients with elevated baseline creatinine (6-14 mg/dL) may have a greater decline in postprocedural hemoglobin when compared with patients having normal serum creatinine levels.[15] In general, consideration of prophylactic plasma or platelet transfusion before thoracentesis should be individualized but is likely unnecessary.[17]

When patient positioning is not optimal (eg, partially recumbent), liver or spleen puncture may occur. Outcomes from this complication are generally favorable if a small-bore needle is used and if the patient is not receiving anticoagulants or does not have a bleeding diathesis.[18]

Removing too much fluid at thoracentesis may be detrimental. A small case series[19] describes consistent and predictable hypoxemia correlating with increasing volume of pleural fluid extracted ($r$ = 0.57; $P$ < .05). In the same study,[19] all patients underwent thoracentesis using an 18-gauge needle with a syringe connected to a stopcock collection system, enabling different volumes of pleural fluid removal. Rapid removal of large fluid volumes with resultant reexpansion pulmonary edema caused hypoxemia that was easily reversed by low-flow oxygen therapy within 24 hours of the procedure.[20] One proposed means for reducing the risk of large-volume (>1L) thoracentesis-associated hypoxemia is to monitor the pleural fluid pressure by manometry as it is removed; a pleural pressure less than −20 cm $H_2O$ has been suggested to be associated with increased risk of reexpansion pulmonary edema.[21] Two studies suggested that greater total changes in pleural pressure (mean [SD], −20 [10] cm $H_2O$; range, 5-43) and not the absolute volume of pleural fluid extracted may be associated with an increased risk of reexpansion pulmonary edema.[22,23] In one study,[23] 1 in 4 patients had an end-of-procedure or closing pleural pressure of less than −20 cm $H_2O$.

### Contraindications to Thoracentesis

Thoracentesis may be deferred for patients with severe hemodynamic instability or respiratory compromise not caused by the effusion itself. Patients receiving positive end-expiratory pressure mechanical ventilation do not have increased risk for pneumothorax compared with patients not receiving such therapy.[24-26] When pneumothorax occurred, risks were greater for tension pneumothorax or subsequent development of bronchopleural fistulae.[27-29]

Among patients receiving therapeutic anticoagulation or with a bleeding diathesis (eg, prolonged prothrombin time or prolonged partial thromboplastin time >1.5× the normal range, platelet count <25 × $10^3$/μL, or serum creatinine >1.7 mg/dL), reversal of the coagulopathy or thrombocytopenia should be individualized to the clinical scenario when performing thoracentesis.[15,16] For instance, when the procedure is to be performed under ultrasound guidance by an experienced operator, the risk of bleeding is so low that testing or reversal of abnormal coagulation may be unnecessary.[30]

The purpose of this systematic review is to identify which tests optimally distinguish between transudative and exudative effusions and to review thoracentesis complications and their preven-



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tion. This study provides a best-practice approach for thoracentesis in adults with pleural effusion based on systematic evidence review and its integration with expert opinion.

## Methods

### Literature Search Strategy

We searched the Cochrane Library (Wiley interface, February 2014), Medline (OVID interface, 1950 to February 2014), and Embase (OVID interface, 1980 to February 2014) to identify relevant studies (eTable 1 in the Supplement).

### Study Selection and Data Extraction

#### Diagnostic Accuracy Studies

We extracted data on the physical characteristics of the effusion (eg, appearance, viscosity), white and red blood cell counts, and the results of commonly available biochemical tests (eg, pleural protein, albumin, cholesterol). Study quality was summarized using a checklist designed for The Rational Clinical Examination series, in which a threshold of 100 patients included in the study was used to distinguish level 1 from level 2 studies.[31] We included only studies for which primary data or appropriate summary statistics were available (see the Supplement for data extraction process).

We retained studies in which the investigators used a microbiological or pathological result to categorize effusions as exudates or transudates. In these studies, the investigators used their clinical knowledge, without regard to Light's criteria, based on the proven etiology of the effusion. For example, patients with effusions secondary to infection, malignancy, or inflammatory conditions were considered to have exudates, whereas patients with heart failure or nephritic syndrome were considered to have transudates. These studies were considered to have the most reliable data because the diagnosis was established independently of the symptoms, signs, and Light's criteria. Several studies used Light's criteria as the reference standard for classifying effusions, rather than the final diagnosis. Because this approach is a common clinical practice, for comparison purposes we show the results from these studies in the Supplement only since studies that test a component of Light's criteria and then used those criteria as a reference standard will be biased toward higher accuracy.

#### Studies of Procedural Methods

The search for this review included only randomized and observational studies of adult patients (aged 18 years or older) undergoing interventions intended to reduce the risk of adverse events (eg, patient discomfort, "dry tap" [no fluid obtained], pneumothorax) at the time of thoracentesis. Interventions and factors of interest included puncture apparatus (eg, needle size) and routine postprocedural chest radiography. Attempt was also made to identify studies that assessed the effect of operator experience. The outcomes of interest included success in obtaining pleural fluid, number of attempts, and incidence of pneumothorax occurring up to 7 days after thoracentesis. The GRADE (Grading of Recommendations Assessment, Development and Evaluation) system was used to rate the overall quality of evidence.[32] More specifically, the overall quality of a study was categorized as high (further research is very unlikely to change confidence in the estimate of effect), moderate (further research is likely to have an important effect on confidence in the estimate of effect and may change the estimate), low (further research is very likely to have an important effect on confidence in the estimate of effect and is likely to change the estimate), or very low (any estimate of effect is uncertain).[32]

### Statistical Methodology

Sensitivity, specificity, and likelihood ratios (LRs) were calculated for studies of test accuracy.[33] If a study contained any zeros in the 2 × 2 table resulting in LR estimates of zero or infinity, 0.5 was added to all counts for that study. For any findings evaluated in only 2 studies, the range for the odds ratio (OR) or LR without a point estimate or 95% CI is reported. For findings evaluated in at least 3 studies, univariate random-effects summary ORs and LRs were calculated using version 1.4 of Meta-DiSc[34] since bivariate measures were similar or failed to converge on a solution.[35] When test results were evaluated at different threshold levels, the data were abstracted at each level and then the optimum threshold was selected based on a balance between the diagnostic OR and the width of its CI.

Review Manager version 5.0.22 was used to calculate summary adverse event rate, summary risk difference and 95% CIs, and pooled ORs and 95% CIs for adverse outcomes of thoracentesis. If 3 or more studies examined the same adverse outcome, heterogeneity was assessed using the $I^2$ statistic to determine the percentage of total variability across studies attributable to heterogeneity rather than chance. Heterogeneity was categorized using published guidelines: low ($I^2$ = 25%-49%), moderate ($I^2$ = 50%-74%), and high ($I^2 \geq 75\%$).[36] To conduct meta-analyses of the risks of pneumothorax, chest tube placement, dry tap, and hemothorax, the proportion of patients in each study who had each complication was converted to the log OR (ie, summary rate) first.[37] The standard error of each log odds, where odds = X/(n − X), X = number of events, and n = total number of patients, was calculated as the square root of $(1/X + 1/[n - X])$. Natural log-transformed odds were pooled using the generic inverse variance method. When there were enough studies to detect possible publication bias (specifically for the analyses of pneumothorax, chest tube placement, dry tap, and hemothorax), funnel plots (scatterplot of standard error of logOR against logOR for each study) were inspected using the Egger regression test.[38] This study used random-effects models, which incorporate between-trial heterogeneity and generally yield wider CIs when heterogeneity is present.

## Results

### Accuracy of Interpretation of Results in Diagnosis of Pleural Effusion

A total of 1914 citations were retrieved for accuracy of pleural fluid analysis in diagnosing a pleural effusion as either transudative or exudative. After applying inclusion and exclusion criteria, 48 were retained (eFigure 1, eTable 2, and eTable 3 in the Supplement).[2-6,39-81] Overall, 4 of these studies were classified as level 1, 1 study was classified as level 2, and 43 studies were classified as level 4 on The Rational Clinical Examination quality scale.[31]

Several biochemical tests were assessed to distinguish between transudative and exudative pleural fluid (**Table 1**). Studies of these tests were difficult to summarize since they included patients with different underlying causes for their pleural effusions (eTable 8, eTable 9, eTable 10 in the Supplement). The most valid studies included patients not selected by underlying disease and for whom the final diagnosis of the cause of the pleural effusion was the



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Table 1. Diagnostic Accuracy for Most Useful Findings for Diagnosis of Pleural Exudate[a]

| Source | Patients, No. | Sensitivity, % (95% CI) | Specificity, % (95% CI) | Positive LR (95% CI) | $I^2$, % | Negative LR (95% CI) | $I^2$, % |
|---|---|---|---|---|---|---|---|
| Pleural cholesterol >55 mg/dL[5,77], range[b] | 379 | 85-94[b] | 95-99[b] | 7.1-250[b] | | 0.07-0.16[b] | |
| Pleural LDH>200 U/L[2,5,66] | 439 | 70 (64-75) | 98 (93-100) | 18 (6.8-46) | 0 | 0.32 (0.27-0.38) | 0 |
| Pleural:serum cholesterol ratio >0.3[4,5,77] | 496 | 93 (90-96) | 94 (90-97) | 14 (5.5-38) | 67 | 0.08 (0.05-0.12) | 0 |
| Pleural:serum LDH ratio >0.6[2,5,6,66,77,81] | 736 | 88 (84-91) | 91 (88-94) | 9.2 (5.9-14) | 22 | 0.14 (0.10-0.20) | 29 |
| Pleural:serum protein ratio >0.5[2,5,6,66,68,77,81] | 753 | 90 (87-93) | 90 (86-93) | 7.0 (2.7-18) | 86 | 0.12 (0.09-0.16) | 0 |
| Combined, ≥1 of Light's criteria[c,2,4,5,72,75,77] | 738 | 97 (95-98) | 85 (81-89) | 5.2 (3.3-8.5) | 68 | 0.04 (0.02-0.11) | 47 |
| Pleural protein >3 g/dL[2,53,57,66] | 270 | 88 (82-92) | 86 (76-93) | 5.1 (2.5-11) | 37 | 0.14 (0.07-0.32) | 67 |
| Pleural LDH>2/3 upper limit of normal[68,77] | 207 | 88-89[b] | 93-100[b] | 1.7-13[b] | | 0.23-0.26[b] | |
| Serum:pleural albumin gradient <1.2 mg/dL[72,74] | 145 | 86-95[b] | 42-100[b] | 1.5-36[b] | | 0.06-0.32[b] | |

Abbreviations: LDH, lactate dehydrogenase; OR, odds ratio.

[a] See eTables 8-10 for results from individual studies in the Supplement.

[b] For findings evaluated in only 2 studies, the range is reported rather than a point estimate with 95% CI. $I^2$ for heterogeneity was determined when there were at least 3 studies.

[c] Light's criteria: (1) ratio of pleural fluid protein to serum protein greater than 0.5; (2) ratio of pleural fluid LDH to serum LDH greater than 0.6; (3) pleural fluid LDH greater than two-thirds the upper limit of normal serum LDH.

reference standard for determining whether the effusion was an exudate or transudate (ie, pathology proven diagnosis). When the final diagnosis was associated with conditions expected to produce a transudate (ie, not a pathology-proven diagnosis but a clinical diagnosis such as congestive heart failure or cirrhosis), the effusion was assumed to be a transudate. Conversely, the effusion was considered to be an exudate when the final diagnosis was an infection, malignancy, or inflammatory process causing the effusion.

Exudative effusions were best diagnosed when the pleural cholesterol was greater than 55 mg/dL (n = 379; sensitivity, 85%-94%; specificity, 95%-99%; LR range, 7.1-250),[5,77] the pleural LDH greater than 200 U/L (n = 439; sensitivity, 70%; 95% CI, 64%-75%; specificity, 98%; 95% CI, 93%-100%; summary positive LR, 18; 95% CI, 6.8-46),[2,5,66] or the ratio of pleural cholesterol to serum cholesterol was greater than 0.3 (n = 496; sensitivity, 93%; 95% CI, 90%-96%; specificity, 94%; 95% CI, 90%-97%; summary positive LR, 14; 95% CI, 5.5-38)[4,5,77] (Table 1).

## Adverse Events

A total of 2665 citations were retrieved regarding interventions intended to reduce the risk associated with performing a thoracentesis. Thirty-seven articles were retained after application of our inclusion and exclusion criteria (see eFigure 2, eTable 4, eTable 5 in the Supplement).[8,12,14,17,30,41,82-112] One study was rated as having high quality, 6 were of moderate quality, and 30 were of low quality, according to the GRADE criteria.[32]

The summary rate for pneumothorax following thoracentesis was 6.0% (95% CI, 4.0%-7.0%; $I^2$ = 95%).* The summary rate for placement of a chest tube following diagnosis of pneumothorax was slightly lower at 2.0% (95% CI, 0.99%-2.9%; $I^2$ = 82%).† Dry tap occurred in 7.4% of procedures (95% CI, 3.8%-13%; $I^2$ = 83%).‡ The risk of significant hemorrhage, defined as either hemothorax (aspiration of bright red blood through needle during procedure) or significant bleeding at puncture site after thoracentesis was 1.0% (95% CI, 0.0%-1.0%; $I^2$ = 49%).[17,30,86,90,100,109,111] Meta-analyses of studies reporting pneumothorax, requirement for chest tube, and hemorrhage showed possible publication bias (*P* value using the Egger regression test was .04, .02, and .02, respectively), where studies with low complication rates may have been preferentially published, but the meta-analysis of studies reporting occurrence of dry tap did not suggest publication bias (*P* value using the Egger regression test was .50). Two studies reported hypotension (range, 0.6%-1.7%)[82,97] and 2 studies reported reexpansion pulmonary edema (new or worsening hypoxemia and chest radiography consistent with edema in a reexpanded lung; range, 0.0%-16%).[82,110]

*References 8, 12, 14, 30, 41, 82-110, 112
†References 8, 41, 82-84, 86, 87, 90, 91, 93-95, 97-99, 103-107
‡References 12, 85, 91, 92, 95, 101, 105, 106, 108

## Factors Affecting Performance of the Procedure

### Procedural Factors

Although OR point estimates were less than 1.0 for several technical aspects of thoracentesis, suggesting fewer associated pneumothorax events, most studies reported broad CIs with an upper bound greater than 1.0 and, therefore, were not statistically significant (**Table 2**). These studies include the use of narrow-gauge compared with larger needles for both diagnostic thoracentesis (summary OR, 0.63; 95% CI, 0.10-4.0; $I^2$ = 76%)[83,96,102,104] and therapeutic thoracentesis (summary OR, 0.69; 95% CI, 0.13-3.7; $I^2$ = 67%),[85,89,104,107] and removing a smaller (500 mL–1 L of pleural fluid) rather than a larger volume (>1 L) of fluid for therapeutic procedures (summary OR, 1.3; 95% CI, 0.63-2.8; $I^2$ = 63%).[82,83,91,98,102]

Several needles designed specifically for thoracentesis were compared with standard needles (20-gauge; 0.91 mm diameter). For example, compared with a standard needle, the Veres needle (2.3 mm diameter)[96] had the lowest OR of all other needle types, suggesting a potential reduced risk for pneumothorax (0.14; 95% CI, 0.02-1.1). For other types of needles, the CIs were much broader: for the Boutin needle (3-mm diameter), OR 1.1 (95% CI, 0.27-4.1)[91]; for the Copes needle (3-mm diameter), OR 0.69 (95% CI, 0.19-2.5)[91]; and for angiocatheter (1.7-mm diameter), OR 0.54 (95% CI, 0.11-2.7).[12] Withdrawal of fluid through a standard needle, as opposed to a plastic catheter, may be associated with an increase in the risk of a dry tap, but the CI was broad (OR, 2.5; 95% CI, 0.51-12) and, therefore, not statistically significant.[12]

The effect of operator experience on the risk of adverse events was unclear, as the data were limited to nonrandomized trials without adequate reporting of confounders. However, no statistically sig-



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Table 2. Factors Affecting Rate of Pneumothorax From Thoracentesis[a]

| Factor | Patients, No. | Comparison | Summary Event Rate, % | Summary Risk Difference (95% CI) | $I^2$, % | Summary OR (95% CI) | $I^2$, % |
|---|---|---|---|---|---|---|---|
| Needle size[83,96,102,104] | 1031 | Smaller than 20 gauge vs larger | 4.5 vs 9.3 | −0.02 (−0.12 to 0.08) | 91 | 0.63 (0.10 to 4.0) | 76 |
| Needle type[12,91,96] | 34 | Standard vs catheter needle (1.7 mm)[12] | 20 vs 31 | −0.12 (−0.41 to 0.18) | | 0.54 (0.11 to 2.7) | |
| | 171 | Standard vs Copes thoracentesis needle (3 mm)[91] | 5.0 vs 7.0 | −0.02 (−0.09 to 0.05) | | 0.69 (0.19 to 2.5) | |
| | 184 | Standard vs Boutin thoracentesis needle (3 mm)[91] | 5.0 vs 4.8 | 0.00 (−0.06 to 0.06) | | 1.1 (0.27 to 4.1) | |
| | 221 | Standard vs Veres thoracentesis needle (2.3 mm)[96] | 1.5 vs 9.2 | −0.08 (−0.13 to −0.02) | | 0.14 (0.02 to 1.1) | |
| Type of procedure[85,89,104,107] | 540 | Diagnostic (<100 mL) vs therapeutic | 4.2 vs 8.5 | −0.03 (−0.14 to 0.09) | 62 | 0.69 (0.13-3.7) | 67 |
| Volume of fluid removed[82,83,91,98,102] | 2050 | Smaller (range, <500mL to <1L) vs larger volume | 4.1 vs 4.9 | −0.01 (−0.04 to 0.02) | 57 | 1.3 (0.63-2.8) | 63 |
| Operator experience[83,91,101,102] | 1303 | Less vs more experience | 4.9 vs 4.6 | 0.01 (−0.07 to 0.02) | 52 | 1.3 (0.53-3.0) | 48 |
| Skin marking of needle insertion site[12,99,104,106] | 724 | Ultrasound-guided skin marking vs localization by physical examination[b] | 6.9 vs 16 | −0.08 (−0.23 to 0.07) | 0 | 0.37 (0.08-1.7) | 74 |
| Procedural ultrasound[94] | 421 | Ultrasound guidance at bedside during the procedure vs no ultrasound guidance[c] | 0.73 vs 1.3 | −0.01 (−0.03 to 0.01) | | 0.55 (0.06-5.3) | |

Abbreviation: OR, odds ratio.

[a] See eTable 6 for results from individual studies in the Supplement.

[b] A radiologist used ultrasound to mark the skin, indicating the optimal site for needle insertion. The needle was then inserted without visualization of the pleural space.

[c] Ultrasound was used to visualize the pleural space and effusion as the needle was inserted.

nificant increase in the risk of adverse events was found when medical students or junior residents performed the procedure (ie, less-experienced operators), relative to senior residents, fellows, or attending physicians (ie, more-experienced operators) (summary OR, 1.3; 95% CI, 0.53-3.0; $I^2$ = 48%).[83,91,101,102]

### Ultrasound Facilitation

Skin marking, performed by a radiologist using ultrasound to localize the pleural fluid and identify the optimal site for needle insertion, was not associated with a statistically significant reduction in pneumothorax (summary OR, 0.37; 95% CI, 0.08-1.7; $I^2$ = 74%, Table 2).[12,99,104,106] Ultrasound-guided bedside needle insertion was also not statistically associated with a decreased risk of pneumothorax (OR, 0.55; 95% CI, 0.06-5.3).[94] However, the benefit of needle insertion with ultrasound guidance may differ between patients with large vs smaller effusions or between those with loculated vs free-flowing effusions.[99] Further studies are needed because currently available data are limited.

Among patients with smaller effusions, needle insertion with ultrasound guidance at the bedside may be associated with a lower risk of a dry tap compared with use of a decubitus chest radiograph for localizing the effusion (OR, 0.23; 95% CI, 0.07-0.72), but there were no statistically significant associations among multiple needle passes (OR, 0.38; 95% CI, 0.13-1.1) or pneumothorax (OR, 0.44; 95% CI, 0.12-1.7).[99] Results were similar for patients with loculated effusions. Needle insertion with ultrasound guidance confers benefits for dry tap (OR, 0.11; 95% CI, 0.02-0.70) but not for multiple passes (OR, 0.20; 95% CI, 0.04-1.2) or pneumothorax (OR, 0.78; 95% CI, 0.06-10).[99] For patients with larger effusions, there may be no benefit of needle insertion with ultrasound guidance, although the broad CIs make this conclusion uncertain: for dry tap, OR 0.40 (95% CI, 0.04-3.9); for multiple passes, OR 1.9 (95% CI, 0.30-12); and for pneumothorax, OR 1.0 (95% CI, 0.33-3.3).[99]

### Symptoms During the Procedure

Table 3 shows that respiratory symptoms (eg, dyspnea or cough) experienced during thoracentesis could cause or be caused by pneumothorax. Regardless of the chronology, such symptoms identify patients who are much more likely to experience pneumothorax (summary OR, 69; 95% CI, 3.2-1491; $I^2$ = 87%).[8,83,89] A similar increase in the odds of pneumothorax was observed with aspiration of air during performance of thoracentesis (summary OR, 52; 95% CI, 13-216; $I^2$ = 39%).[8,83,93] Despite a broad CI, the summary OR suggested that the risk of pneumothorax may increase with an increasing number of needle passes; however, it did not reach statistical significance (summary OR, 2.3; 95% CI, 0.55-9.8; $I^2$ = 70%).[83,91,93]

## Limitations

Reviewing the characteristics distinguishing exudative from transudative effusions was difficult because many studies did not use an appropriate reference standard. The physician must make a clinical diagnosis of the cause of the effusion (eg, malignancy, congestive heart failure, or pneumonia). Therefore, studies that did not investigate beyond assessment of Light's criteria (which represent an intermediate end point rather than the underlying diagnosis), may have incorrectly categorized effusions as being transudate or exudate, particularly if it was assumed that the character of the fluid rules in or rules out a particular diagnosis.

Substantial heterogeneity was found in how thoracentesis procedures were performed. Furthermore, this study was also unable to fully explore possible differences in the likelihood of adverse outcomes from therapeutic procedures vs diagnostic ones. For example, because the effusions are larger, pneumothorax rates may be lower for therapeutic procedures than for diagnostic procedures (although patients undergoing therapeutic thoracentesis would most certainly have a greater risk of reexpansion pulmonary edema).[110] Pneumothorax rates in this analysis were lower than



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Table 3. Thoracentesis-Related Factors Identifying Patients at Risk for Pneumothorax

| Source | Patients, No. | Summary Event Rate, % | Summary Risk Difference (95% CI) | $I^2$, % | OR (95% CI) for Pneumothorax | $I^2$, % |
|---|---|---|---|---|---|---|
| Patient experienced symptoms during procedure | | | | | | |
| Aleman,[83] 1999 | 506 | 70 (symptomatic) vs 0.20 (asymptomatic) | 0.33 (0.05 to 0.61) | | 1 141 (129 to 10 000) | |
| Capizzi,[89] 1998 | 104 | 75 (symptomatic) vs 5.8 (asymptomatic) | 0.69 (0.17 to 1.1) | | 49 (4.4 to 545) | |
| Collins,[8] 1987 | 129 | 42 (symptomatic) vs 8.5 (asymptomatic) | 0.70 (0.50 to 0.90) | | 7.6 (2.1 to 29) | |
| Summary OR | | | 0.57 (0.32 to 0.82) | 56 | 69 (3.2 to 1491) | 87 |
| Air aspirated during procedure | | | | | | |
| Aleman,[83] 1999 | 506 | 80 (air aspirated) vs 3.2 (no air aspirated) | 0.77 (0.42 to 1.1) | | 122 (12.9 to 1157) | |
| Doyle,[93] 1996 | 174 | 31 (air aspirated) vs 2.5 (no air aspirated) | 0.29 (0.06 to 0.52) | | 18 (4.1 to 75) | |
| Collins,[8] 1987 | 129 | 90 (air aspirated) vs 6.5 (no air aspirated) | 0.83 (0.64 to 1.0) | | 1.25 (0.16 to 9.8) | |
| Summary OR | | | 0.63 (0.23 to 1.0) | 87 | 52 (13 to 216) | 39 |
| More vs fewer needle passes | | | | | | |
| Doyle,[93] 1996 | 174 | 14 (more passes) vs 2.3 (fewer passes) | 0.12 (0.01 to 0.22) | | 6.9 (1.6 to 29) | |
| Colt,[91] 1999 | 255 | 13 (more passes) vs 4.2 (fewer passes) | 0.08 (−0.02 to 0.19) | | 3.3 (1.0-10) | |
| Aleman,[83] 1999 | 506 | 2.0 (more passes) vs 3.9 (fewer passes) | −0.02 (−0.05 to 0.01) | | 0.51 (0.12-2.3) | |
| Summary OR | | | 0.05 (−0.07 to 0.17) | 84 | 2.3 (0.55-9.8) | 70 |

Abbreviation: OR, odds ratio.

expected,[18] which might be attributable to publication bias. Published studies might not reflect typical clinical practice and nonrandomized studies may generate misleading results (compared with randomized trials of interventions to optimize the safety of thoracentesis), even when intervention and control groups appear to have similar baseline characteristics.[113-116] Similarly, few studies assessed location of needle insertion.

This study did not find a statistically significant benefit of ultrasound marking to localize pleural fluid for the needle insertion site for pneumothorax rates. There also was no statistically significant benefit related to thoracentesis performed by either a radiologist (even when the remainder of the procedure is performed by the most capable physician after the site has been located) or a practitioner experienced in performing ultrasound-guided thoracentesis is at the bedside.[13] One reason a benefit might not have been observed may be the absence of randomized trials. The trials in this review included results from before-and-after observational studies, which tend to overestimate the effect of an intervention because of secular trends, even after application of standard methods to adjust for differences. Furthermore, the absence of subgroup analyses by ultrasound operators limits the generalizability of findings because it is unclear whether all patients would benefit equally (eg, regardless of the size of the effusion). Although the point estimates suggest that ultrasound guidance may be associated with a lower risk of pneumothorax, the CIs from extant studies suggest the possibility that this risk might be increased. The data do not allow determination of whether ultrasound is preferentially used for patients at greater risk for a complication or whether less experienced physicians request ultrasound more than experienced physicians do.

All included studies of ultrasound marking had radiologists or radiology residents in operant roles. As such, this review cannot firmly say that nonradiologist markings would be similarly effective simply because these studies do not exist. However, we acknowledge that many nonradiologist physicians are comfortable identifying an effusion with ultrasound guidance. What is not known is whether their comfort is justified by their own thoracentesis results, given the CIs that suggest the possibility of harm.

Chest radiography may not be required routinely[101,117] but should be done if the patient experiences symptoms (eg, dyspnea or cough) or air is aspirated during the procedure to rule out pneumothorax.[83,89,93] The risk of pneumothorax may have been underestimated because not all articles meeting inclusion criteria required chest radiography immediately after routine thoracentesis; however, this review did not find any evidence to support this practice.

## How Thoracentesis Should Be Performed

The following description of the method to perform thoracentesis considers the best available evidence; textbooks were used to fill gaps not supported by trial evidence, and opinions from experts.

The procedure and its risks should be explained to the patient and informed consent obtained (**Figure**).[118]

Have the patient sit on the edge of the bed, leaning forward, with arms resting on a bedside table. If the patient is unable to sit upright, the lateral recumbent or supine position is acceptable.[117]

The needle should be inserted 1 or 2 intercostal spaces below the level of the effusion, 5 to 10 cm lateral to the spine.[117] To avoid intra-abdominal injury, the needle should not be inserted below the ninth rib.

The operator should then mark the appropriate site, prepare the skin with antiseptic solution (0.05% chlorhexidine or 10% povidine-iodine solution), and apply a sterile drape.[117]

The overlying epidermis of the superior edge of the rib that lies below the selected intercostal space should be anesthetized using a small (25-gauge) needle.

A larger (20-gauge) needle should then be inserted and "walked" along the superior edge of the rib, alternately injecting anesthetic (1% or 2% lidocaine) and pulling back on the plunger every few millimeters to rule out intravascular placement and to check for proper intrapleural placement.

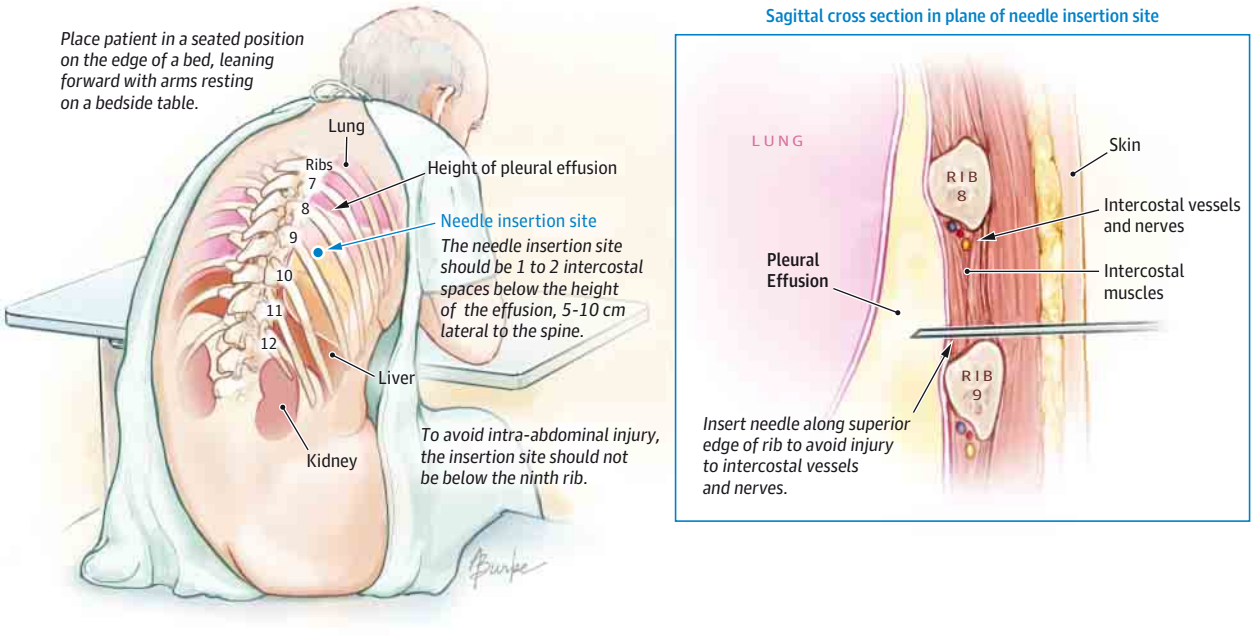
The needle should not touch the inferior surface of the rib so as to avoid injury to the intercostal nerves and vessels. Once pleural fluid is aspirated, additional lidocaine should be injected to anesthetize the highly sensitive parietal pleura.[117]



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Figure. Patient Position and Needle Placement When Performing a Thoracentesis




Once pleural fluid is obtained, the needle should no longer be advanced, to avoid puncture of the lung. Additional lidocaine should be injected to anesthetize the highly sensitive parietal pleura.

After removal of the needle, the open hub of the catheter should be covered with a gloved finger to prevent the entry of air into the pleural cavity and a 3-way stopcock attached to the catheter hub.[117]

With the stopcock open to the patient and the syringe, aspirate a minimum of 10 mL of pleural fluid for diagnostic analysis and then close the stopcock to the patient. If additional fluid is to be removed for therapeutic purposes, one end of the high-pressure drainage tubing can be attached to the third port of the stopcock and the other end to a large evacuated container.[117] The stopcock should then be opened to the patient and the container, and the fluid should be allowed to drain. No more than 1 L of fluid should be removed during a therapeutic thoracentesis.[82,83,91,98,102]

When the procedure is complete, the needle or catheter should be removed while the patient holds his/her breath or forcibly contracts the abdominal muscles at end expiration. The site should be covered with an occlusive dressing and the remaining antiseptic solution removed from the skin.[117] All needles should be placed in appropriate safety containers.

Chest radiography is not routinely required after thoracentesis; however, it should be performed if the patient experienced symptoms such as dyspnea or cough during the procedure, or if air was aspirated.[83,89,93,117,119]

## How Thoracentesis Should Be Taught or Learned

On the basis of a recent survey of internal medicine residency program directors in the United States, it was recommended that a mean of 5 thoracentesis procedures be performed (interquartile range [IQR], 3-10 procedures) to attain procedural competency and that a mean of 4 procedures (IQR, 2-5 procedures) be performed every year to maintain competency.[120] The evidence reviewed in this article indicates targets that could help clinicians assess the quality of their thoracentesis performance. Successful thoracentesis is indicated by obtaining sufficient pleural fluid for analysis on the first attempt and by achieving a rate of procedure-associated pneumothorax less than 6%.[13] However, because this figure represents the average risk of events, it cannot be considered a benchmark.[121] As such, all physicians should consider keeping personal training logs to assess their own adverse event rates.

To date, there is little evidence to guide the teaching of this procedure. Simulators[122] and procedural checklists[123,124] have been developed but have not been rigorously evaluated. One study used a pretest-posttest observational checklist to evaluate a 2-hour educational session designed to enhance procedural performance skills.[125] Using simulation technology, resident performance improved by 71% with deliberate practice.[125] More structured curricula for procedural skills training have been developed to minimize the variation in students' ability and comfort level that may arise because of random and unpredictable acquisition of basic skills through ward teaching.[126,127] The lack of data on use of bedside ultrasonography to guide needle insertion prevents us from providing guidance on how to teach this aspect of the procedure. Physicians may, however, want to consider procedural training on mannequins to determine the adequacy of their skills at thoracic ultrasound prior to clinical practice.[128] Both the American College of Emergency Physicians[129] and the American College of Surgeons[130] strongly support the use of ultrasound for thoracentesis through their policies on scope of practice, training, and maintenance of competency.[131] Further research is required to determine how best to assess clinical competency in bedside ultrasound techniques.



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## Scenario Resolution

The patient has heart failure (which would cause a transudative effusion), but the patient may also have pneumonia (which could cause an exudative effusion). The treating physician's clinical judgment is that there is a 50% chance of an exudative effusion. The patient's consent to perform thoracentesis should be obtained. The fluid and a sample of the patient's serum should be sent for measurement of LDH and protein; and cholesterol levels should be requested, knowing that the latter may offer better operating characteristics for distinguishing a transudative from an exudative effusion. Pleural LDH is 220 U/L (LR = 18 for an exudate), pleural protein is 54 g/L (LR = 5.1 for an exudate), pleural cholesterol is 56 mg/dL (LR range, 7.1-250), serum LDH is 342 U/L (pleural:serum LDH ratio >0.6; LR, 9.2 for an exudate), serum protein is 35g/L (pleural:serum protein ratio >0.5; LR, 7.0 for an exudate), and serum cholesterol is 156 mg/dL (pleural:serum cholesterol ratio >0.3; LR, 14 for an exudate). All of these results consistently favor an exudate and lead to initiating therapy with appropriate antibiotics and drainage of the parapneumonic effusion. Using the lowest LR for pleural protein (LR 5.1), the probability of an exudate is greater than 84%.

## Clinical Bottom Line

According to a reference standard for the final clinical diagnosis, if the effusion meets none of Light's criteria, it is transudative. If the effusion meets Light's criteria or if any of the following results are obtained, the effusion is most likely exudative: pleural cholesterol greater than 55 mg/dL (LR range, 7.1-250); pleural LDH greater than 200 U/L (LR, 18; 95% CI, 6.8-46); and ratio of pleural cholesterol to serum cholesterol greater than 0.3 (LR, 14; 95% CI, 5.5-38).

No specific precaution has been definitively shown to reduce the risk of pneumothorax. However, the following may be helpful: use of a small-gauge needle (at least 20 gauge); and removal of less than 1 L of pleural fluid at a time.

Randomized trials are needed to evaluate whether ultrasound marking by a radiologist or needle insertion under ultrasound guidance at the bedside is needed for all effusions and for those of all levels of operator experience because current literature consists of primarily nonrandomized data with inherent methodological limitations.

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REFERENCES

1. Maskell NA, Butland RJA; Pleural Diseases Group, Standards of Care Committee, British Thoracic Society. BTS guidelines for the investigation of a unilateral pleural effusion in adults. *Thorax*. 2003; 58(suppl 2):ii8-ii17.

2. Light RW, Macgregor MI, Luchsinger PC, Ball WC Jr. Pleural effusions: the diagnostic separation of transudates and exudates. *Ann Intern Med*. 1972;77 (4):507-513.

3. Garcia-Pachon E, Padilla-Navas I, Sanchez JF, Jimenez B, Custardoy J. Pleural fluid to serum cholinesterase ratio for the separation of transudates and exudates. *Chest*. 1996;110(1):97-101.

4. Hamm H, Brohan U, Bohmer R, Missmahl HP. Cholesterol in pleural effusions. *Chest*. 1987;92(2): 296-302.

5. Valdés L, Pose A, Suàrez J, et al. Cholesterol: a useful parameter for distinguishing between pleural exudates and transudates. *Chest*. 1991;99 (5):1097-1102.

6. Meisel S, Shamiss A, Thaler M, et al. Pleural fluid to serum bilirubin concentration ratio for the separation of transudates from exudates. *Chest*. 1990;98(1):141-144.

7. Porcel JM. Pearls and myths in pleural fluid analysis. *Respirology*. 2011;16(1):44-52.

8. Collins TR, Sahn SA. Thoracocentesis: clinical value, complications, technical problems, and patient experience. *Chest*. 1987;91(6):817-822.

9. Heffner JE, Sahn SA. Abdominal hemorrhage after perforation of a diaphragmatic artery during thoracentesis. *Arch Intern Med*. 1981;141(9):1238.

10. Stewart BN, Block AJ. Subcutaneous implantation of cancer following thoracentesis. *Chest*. 1974;66(4):456-457. doi:10.1378/chest.66.4.456

11. Sue DY, Lam K. Retention of catheter fragment after thoracentesis. *Postgrad Med*. 1982;72(1): 101-102-105-106.

12. Grogan DR, Irwin RS, Channick R, et al. Complications associated with thoracentesis. *Arch Intern Med*. 1990;150(4):873-877.

13. Gordon CE, Feller-Kopman D, Balk EM, Smetana GW. Pneumothorax following thoracentesis. *Arch Intern Med*. 2010;170(4):332-339.

14. Mercaldi CJ, Lanes SF. Ultrasound guidance decreases complications and improves the cost of care among patients undergoing thoracentesis and paracentesis. *Chest*. 2013;143(2):532-538.

15. McVay PA, Toy PTCY. Lack of increased bleeding after paracentesis and thoracentesis in patients with mild coagulation abnormalities. *Transfusion*. 1991;31(2):164-171.

16. Patel MD, Joshi SD. Abnormal preprocedural international normalized ratio and platelet counts are not associated with increased bleeding complications after ultrasound-guided thoracentesis. *AJR Am J Roentgenol*. 2011;197(1): W164-W1688.

17. Hibbert RM, Atwell TD, Lekah A, et al. Safety of ultrasound-guided thoracentesis in patients with abnormal preprocedural coagulation parameters. *Chest*. 2013;144(2):456-463.

18. Simon RR, Brenner BE, eds. *Procedures and Techniques in Emergency Medicine*. Baltimore, MD: Wiliams & WIlkins; 1982.

19. Brandstetter RD, Cohen RP. Hypoxemia after thoracentesis. *JAMA*. 1979;242(10):1060-1061.

20. Brandstetter RD, Gyetko MR, Thomas JD, Fellows CL. Hypoxemia following thoracentesis. *Heart Lung*. 1982;11(3):216.

21. Light RW, Jenkinson SG, Minh VD, George RB. Observations on pleural fluid pressures as fluid is



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withdrawn during thoracentesis. *Am Rev Respir Dis*. 1980;121(5):799-804.

**22**. Feller-Kopman D, Berkowitz D, Boiselle P, Ernst A. Large-volume thoracentesis and the risk of reexpansion pulmonary edema. *Ann Thorac Surg*. 2007;84(5):1656-1661.

**23**. Feller-Kopman D, Walkey A, Berkowitz D, Ernst A. The relationship of pleural pressure to symptom development during therapeutic thoracentesis. *Chest*. 2006;129(6):1556-1560.

**24**. Godwin JE, Sahn SA. Thoracentesis: a safe procedure in mechanically ventilated patients. *Ann Intern Med*. 1990;113(10):800-802.

**25**. McCartney JP, Adams JW II, Hazard PB. Safety of thoracentesis in mechanically ventilated patients. *Chest*. 1993;103(6):1920-1921.

**26**. Mayo PH, Goltz HR, Tafreshi M, Doelken P. Safety of ultrasound-guided thoracentesis in patients receiving mechanical ventilation. *Chest*. 2004;125(3):1059-1062.

**27**. Petersen GW, Baier H. Incidence of pulmonary barotrauma in a medical ICU. *Crit Care Med*. 1983;11 (2):67-69.

**28**. Rohlfing BM, Webb WR, Schlobohm RM. Ventilator-related extra-alveolar air in adults. *Radiology*. 1976;121(1):25-31.

**29**. Zimmerman JE, Dunbar BS, Klingenmaier CH. Management of subcutaneous emphysema, pneumomediastinum, and pneumothorax during respirator therapy. *Critical Care Medicine*. 1975;3(2): 69-73.

**30**. Patel PA, Ernst FR, Gunnarsson CL. Ultrasonography guidance reduces complications and costs associated with thoracentesis procedures. *J Clin Ultrasound*. 2012;40(3):135-141.

**31**. Simel DL, Rennie D, eds. *Rational Clinical Examination. Update: Primer on Precision and Accuracy*. New York, NY: McGraw-Hill; 2008.

**32**. Guyatt GH, Oxman AD, Schünemann HJ, Tugwell P, Knottnerus A. GRADE guidelines: a new series of articles in the *Journal of Clinical Epidemiology*. *J Clin Epidemiol*. 2011;64(4):380-382.

**33**. Simel DL, Samsa GP, Matchar DB. Likelihood ratios with confidence. *J Clin Epidemiol*. 1991;44(8): 763-770.

**34**. Zamora J, Abraira V, Muriel A, Khan K, Coomarasamy A. Meta-DiSc: a software for meta-analysis of test accuracy data. *BMC Med Res Methodol*. 2006;6(31):31.

**35**. Simel DL, Bossuyt PMM. Differences between univariate and bivariate models for summarizing diagnostic accuracy may not be large. *J Clin Epidemiol*. 2009;62(12):1292-1300.

**36**. Higgins JP, Thompson SG, Deeks JJ, Altman DG. Measuring inconsistency in meta-analyses. *BMJ*. 2003;327(7414):557-560.

**37**. Goligher EC, Leis JA, Fowler RA, Pinto R, Adhikari NKJ, Ferguson ND. Utility and safety of draining pleural effusions in mechanically ventilated patients. *Crit Care*. 2011;15(1):R46.

**38**. Egger M, Davey Smith G, Schneider M, Minder C. Bias in meta-analysis detected by a simple, graphical test. *BMJ*. 1997;315(7109):629-634.

**39**. Atalay F, Ernam D, Hasanoglu HC, Karalezli A, Kaplan O. Pleural adenosine deaminase in the separation of transudative and exudative pleural effusions. *Clin Biochem*. 2005;38(12):1066-1070.

**40**. Buckley O, Benfayed W, Geoghegan T, et al. Thoracocentesis for potential malignancy. *Hong Kong J Radiol*. 2008;11(2):72-75.

**41**. Escudero Bueno C, García Clemente M, Cuesta Castro B, et al. Cytologic and bacteriologic analysis of fluid and pleural biopsy specimens with Cope's needle. *Arch Intern Med*. 1990;150(6):1190-1194.

**42**. Cardozo P. A critical evaluation of 3000 cytologic analyses of pleural fluid, ascitic fluid and pericardial fluid. *Acta Cytol*. 1966;10:455-460.

**43**. Carr DT, Power MH. Clinical value of measurements of concentration of protein in pleural fluid. *N Engl J Med*. 1958;259(19):926-927.

**44**. Jiménez Castro D, Díaz Nuevo G, Sueiro A, et al. Pleural fluid parameters identifying complicated parapneumonic effusions. *Respiration*. 2005;72(4):357-364.

**45**. Cincin A, Abul Y, Ozben B, et al. Pleural fluid amino-terminal brain natriuretic peptide in patients with pleural effusions. *Respir Care*. 2013;58(2):313-319.

**46**. Costa M, Quiroga T, Cruz E. Measurement of pleural fluid cholesterol and lactate dehydrogenase. *Chest*. 1995;108(5):1260-1263.

**47**. Dines DE, Pierre RV, Franzen SJ. The value of cells in the pleural fluid in the differential diagnosis. *Mayo Clin Proc*. 1975;50(10):571-572.

**48**. Gázquez I, Porcel JM, Vives M, Vicente de Vera MC, Rubio M, Rivas MC. Comparative analysis of Light's criteria and other biochemical parameters for distinguishing transudates from exudates. *Respir Med*. 1998;92(5):762-765.

**49**. Good JT Jr, Taryle DA, Maulitz RM, Kaplan RL, Sahn SA. The diagnostic value of pleural fluid pH. *Chest*. 1980;78(1):55-59.

**50**. Gotsman I, Fridlender Z, Meirovitz A, et al. The evaluation of pleural effusions in patients with heart failure. *Am J Med*. 2001;111(5):375-378.

**51**. Hackbarth JS, Murata K, Reilly WM, Algeciras-Schimnich A. Performance of CEA and CA19-9 in identifying pleural effusions caused by specific malignancies. *Clin Biochem*. 2010;43(13-14): 1051-1055.

**52**. Han CH, Choi JE, Chung JH. Clinical utility of pleural fluid NT-pro brain natriuretic peptide (NT-proBNP) in patients with pleural effusions. *Intern Med*. 2008;47(19):1669-1674.

**53**. Iqbal M, Jaffery T, Shah SH. Isolated pleural fluid lactic dehydrogenase level. *J Ayub Med Coll*. 2002;14(2):2-5.

**54**. James P, Gupta R, Christopher DJ, Balamugesh T. Evaluation of the diagnostic yield and safety of closed pleural biopsy in the diagnosis of pleural effusion. *Indian J Tuberc*. 2010;57(1):19-24.

**55**. Krenke R, Nasilowski J, Korczynski P, et al. Incidence and aetiology of eosinophilic pleural effusion. *Eur Respir J*. 2009;34(5):1111-1117.

**56**. Lakhotia M, Mehta SR, Mathur D, Baid CS, Varma AR. Diagnostic significance of pleural fluid eosinophilia during initial thoracocentesis. *Indian J Chest Dis Allied Sci*. 1989;31(4):259-264.

**57**. Leuallen EC, Carr DT. Pleural effusion; a statistical study of 436 patients. *N Engl J Med*. 1955; 252(3):79-83.

**58**. Light RW, Ball WC Jr. Lactate dehydrogenase isoenzymes in pleural effusions. *Am Rev Respir Dis*. 1973;108(3):660-664.

**59**. Light RW, Erozan YS, Ball WC Jr. Cells in pleural fluid. *Arch Intern Med*. 1973;132(6):854-860.

**60**. Light RW, MacGregor MI, Ball WC Jr, Luchsinger PC. Diagnostic significance of pleural fluid pH and PCO2. *Chest*. 1973;64(5):591-596.

**61**. Light RW, Ball WC Jr. Glucose and amylase in pleural effusions. *JAMA*. 1973;225(3):257-260.

**62**. Marinho FCA, Vargas FS, Fabri J Jr, et al. Clinical usefulness of B-type natriuretic peptide in the diagnosis of pleural effusions due to heart failure. *Respirology*. 2011;16(3):495-499.

**63**. Martínez-García MA, Cases-Viedma E, Cordero-Rodríguez PJ, et al. Diagnostic utility of eosinophils in the pleural fluid. *Eur Respir J*. 2000; 15(1):166-169.

**64**. Ong KC, Indumathi V, Poh WT, Ong YY. The diagnostic yield of pleural fluid cytology in malignant pleural effusions. *Singapore Med J*. 2000;41(1):19-23.

**65**. Ozcakar B, Martinez CH, Morice RC, et al. Does pleural fluid appearance really matter? *J Cardiothorac Surg*. 2010;5:63.

**66**. Paradis IL, Caldwell EJ. Diagnostic approach to a pleural effusion. *J Maine Med Assoc*. 1977;68(10): 378-382.

**67**. Porcel JM, Esquerda A, Bielsa S. Diagnostic performance of adenosine deaminase activity in pleural fluid. *Eur J Intern Med*. 2010;21(5):419-423.

**68**. Porcel JM, Madroñero AB, Pardina M, et al. Analysis of pleural effusions in acute pulmonary embolism. *Respirology*. 2007;12(2):234-239.

**69**. Riantawan P, Sangsayan P, Bangpattanasiri K, Rojanaraweewong P. Limited additive value of pleural fluid carcinoembryonic antigen level in malignant pleural effusion. *Respiration*. 2000;67(1): 24-29.

**70**. Romero Candeira S, Hernández Blasco L, Soler MJ, Muñoz A, Aranda I. Biochemical and cytologic characteristics of pleural effusions secondary to pulmonary embolism. *Chest*. 2002;121(2):465-469.

**71**. Romero-Candeira S, Hernández L, Romero-Brufao S, Orts D, Fernández C, Martín C. Is it meaningful to use biochemical parameters to discriminate between transudative and exudative pleural effusions? *Chest*. 2002;122(5):1524-1529.

**72**. Roth BJ, O'Meara TF, Cragun WH. The serum-effusion albumin gradient in the evaluation of pleural effusions. *Chest*. 1990;98(3):546-549.

**73**. Rubins JB, Rubins HB. Etiology and prognostic significance of eosinophilic pleural effusions. *Chest*. 1996;110(5):1271-1274.

**74**. Sangsayunh P, Saejueng B. Benefit of serum-effusion albumin gradient in congestive heart failure patients. *J Med Assoc Thai*. 2012;95 (suppl 8):S6-S10.

**75**. Scheurich JW, Keuer SP, Graham DY. Pleural effusion: comparison of clinical judgment and Light's criteria in determining the cause. *South Med J*. 1989;82(12):1487-1491.

**76**. Shen Y, Yang T, Jia L, et al. A potential role for D-dimer in the diagnosis of tuberculous pleural effusion. *Eur Rev Med Pharmacol Sci*. 2013;17(2): 201-205.

**77**. Gil Suay V, Martínez Moragón E, Cases Viedma E, et al. Pleural cholesterol in differentiating transudates and exudates. *Respiration*. 1995;62(2): 57-63.



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**78**. Swiderek J, Morcos S, Donthireddy V, et al. Prospective study to determine the volume of pleural fluid required to diagnose malignancy. *Chest*. 2010;137(1):68-73.

**79**. Tavana S, Tavakoli H, Hashemzadeh M, Nadi E. Specific gravity of pleural fluid determined by refractometer to discriminate exudates and transudates. *Res J Med Sci*. 2009;3(3):91-94.

**80**. Villena V, López-Encuentra A, García-Luján R, et al. Clinical implications of appearance of pleural fluid at thoracentesis. *Chest*. 2004;125(1):156-159.

**81**. Yetkin O, Tek I, Kaya A, Ciledag A, Numanoglu N. A simple laboratory measurement for discrimination of transudative and exudative pleural effusion. *Respir Med*. 2006;100(7):1286-1290.

**82**. Abunasser J, Brown R. Safety of large-volume thoracentesis. *Conn Med*. 2010;74(1):23-26.

**83**. Alemán C, Alegre J, Armadans L, et al. The value of chest roentgenography in the diagnosis of pneumothorax after thoracentesis. *Am J Med*. 1999;107(4):340-343.

**84**. Barnes TW, Morgenthaler TI, Olson EJ, Hesley GK, Decker PA, Ryu JH. Sonographically guided thoracentesis and rate of pneumothorax. *J Clin Ultrasound*. 2005;33(9):442-446.

**85**. Bartter T, Mayo PD, Pratter MR, Santarelli RJ, Leeds WM, Akers SM. Lower risk and higher yield for thoracentesis when performed by experienced operators. *Chest*. 1993;103(6):1873-1876.

**86**. Bass J, White DA. Thoracentesis in patients with hematologic malignancy. *Chest*. 2005;127(6): 2101-2105.

**87**. Boland GW, Gazelle GS, Girard MJ, Mueller PR. Asymptomatic hydropneumothorax after therapeutic thoracentesis for malignant pleural effusions. *AJR Am J Roentgenol*. 1998;170(4):943-946.

**88**. Brandstetter RD, Karetzky M, Rastogi R, Lolis JD. Pneumothorax after thoracentesis in chronic obstructive pulmonary disease. *Heart Lung*. 1994; 23(1):67-70.

**89**. Capizzi SA, Prakash UB. Chest roentgenography after outpatient thoracentesis. *Mayo Clin Proc*. 1998;73(10):948-950.

**90**. Clark SJ, Vanselow C, Colt HG. Use of the reusable boutin pleural needle for thoracentesis. *J Bronchology*. 1999;6(3):207-210. doi:10.1097 /00128594-199907000-00015

**91**. Colt HG, Brewer N, Barbur E. Evaluation of patient-related and procedure-related factors contributing to pneumothorax following thoracentesis. *Chest*. 1999;116(1):134-138.

**92**. Diacon AH, Brutsche MH, Solèr M. Accuracy of pleural puncture sites. *Chest*. 2003;123(2):436-441.

**93**. Doyle JJ, Hnatiuk OW, Torrington KG, Slade AR, Howard RS. Necessity of routine chest roentgenography after thoracentesis. *Ann Intern Med*. 1996;124(9):816-820.

**94**. Duncan DR, Morgenthaler TI, Ryu JH, Daniels CE. Reducing iatrogenic risk in thoracentesis. *Chest*. 2009;135(5):1315-1320.

**95**. Grodzin CJ, Balk RA. Indwelling small pleural catheter needle thoracentesis in the management of large pleural effusions. *Chest*. 1997;111(4):981-988.

**96**. Jenkins DW Jr, McKinney MK, Szpak MW, Booker JL Jr. Veres needle in the pleural space. *South Med J*. 1983;76(11):1383-1385.

**97**. Jones PW, Moyers JP, Rogers JT, Rodriguez RM, Lee YCG, Light RW. Ultrasound-guided thoracentesis. *Chest*. 2003;123(2):418-423.

**98**. Josephson T, Nordenskjold CA, Larsson J, Rosenberg LU, Kaijser M. Amount drained at ultrasound-guided thoracentesis and risk of pneumothorax. *Acta Radiol*. 2009;50(1):42-47.

**99**. Kohan JM, Poe RH, Israel RH, et al. Value of chest ultrasonography versus decubitus roentgenography for thoracentesis. *Am Rev Respir Dis*. 1986;133(6):1124-1126.

**100**. Lisi M, Cameli M, Mondillo S, et al. Incremental value of pocket-sized imaging device for bedside diagnosis of unilateral pleural effusions and ultrasound-guided thoracentesis. *Interact Cardiovasc Thorac Surg*. 2012;15(4):596-601.

**101**. Petersen WG, Zimmerman R. Limited utility of chest radiograph after thoracentesis. *Chest*. 2000; 117(4):1038-1042.

**102**. Pihlajamaa K, Bode MK, Puumalainen T, Lehtimaki A, Marjelund S, Tikkakoski T. Pneumothorax and the value of chest radiography after ultrasound-guided thoracocentesis. *Acta Radiol*. 2004;45(8):828-832.

**103**. Puchalski JT, Argento AC, Murphy TE, et al. Etiologies of bilateral pleural effusions. *Respir Med*. 2013;107(2):284-291.

**104**. Raptopoulos V, Davis LM, Lee G, Umali C, Lew R, Irwin RS. Factors affecting the development of pneumothorax associated with thoracentesis. *AJR Am J Roentgenol*. 1991;156(5):917-920.

**105**. Rasmussen OS, Boris P. Ultrasound guided puncture of pleural fluid collections and superficial thoracic masses. *Eur J Radiol*. 1989;9(2):91-92.

**106**. Seneff MG, Corwin RW, Gold LH, Irwin RS. Complications associated with thoracocentesis. *Chest*. 1986;90(1):97-100.

**107**. Villena V, Lopez-Encuentra A, Pozo F, De-Pablo A, Martin-Escribano P. Measurement of pleural pressure during therapeutic thoracentesis. *Am J Respir Criti Care Med*. 2000;162(4I):1534-1538.

**108**. Weingardt JP, Guico RR, Nemcek AA Jr, Li YP, Chiu ST. Ultrasound findings following failed, clinically directed thoracenteses. *J Clin Ultrasound*. 1994;22(7):419-426.

**109**. Zalt MB, Bechara RI, Parks C, Berkowitz DM. Effect of routine clopidogrel use on bleeding complications after ultrasound-guided thoracentesis. *J Bronchology Interv Pulmonol*. 2012; 19(4):284-287.

**110**. Zanforlin A, Gavelli G, Oboldi D, Galletti S. Ultrasound-guided thoracenthesis: the V-point as a site for optimal drainage positioning. *Eur Rev Med Pharmacol Sci*. 2013;17(1):25-28.

**111**. Puchalski JT, Argento AC, Murphy TE, Araujo KLB, Pisani MA. The safety of thoracentesis in patients with uncorrected bleeding risk. *Ann Am Thorac Soc*. 2013;10(4):336-341.

**112**. Soldati G, Smargiassi A, Inchingolo R, et al. Ultrasound-guided pleural puncture in supine or recumbent lateral position: feasibility study. *Multidisciplinary Respiratory Medicine*. 2013;8(3).

**113**. Golder S, Loke Y, McIntosh HM. Poor reporting and inadequate searches were apparent in systematic reviews of adverse effects. *J Clin Epidemiol*. 2008;61(5):440-448.

**114**. Golder S, Loke Y, McIntosh HM. Room for improvement? A survey of the methods used in systematic reviews of adverse effects. *BMC Med Res Methodol*. 2006;6:3.

**115**. Audigé L, Bhandari M, Griffin D, Middleton P, Reeves BC. Systematic reviews of nonrandomized clinical studies in the orthopaedic literature. *Clin Orthop Relat Res*. 2004;(427):249-257.

**116**. Deeks JJ, Dinnes J, D'Amico R, et al Evaluating non-randomised intervention studies. *Health Technol Assess*. 2003;7(27):1-173.

**117**. Thomsen TW, DeLaPena J, Setnik GS. Videos in clinical medicine: thoracentesis. *N Engl J Med*. 2006;355(15):e16.

**118**. Manthous CA, DeGirolamo A, Haddad C, Amoateng-Adjepong Y. Informed consent for medical procedures: local and national practices. *Chest*. 2003;124(5):1978-1984.

**119**. Feller-Kopman D. Therapeutic thoracentesis: the role of ultrasound and pleural manometry. *Curr Opin Pulm Med*. 2007;13(4):312-318.

**120**. Wigton RS, Blank LL, Nicolas JA, Tape TG. Procedural skills training in internal medicine residencies. *Ann Intern Med*. 1989;111(11):932-938.

**121**. Kiefe CI, Weissman NW, Allison JJ, Farmer R, Weaver M, Williams OD. Identifying achievable benchmarks of care: concepts and methodology. *Int J Qual Health Care*. 1998;10(5):443-447.

**122**. Jiang G, Chen H, Wang S, et al. Learning curves and long-term outcome of simulation-based thoracentesis training for medical students. *BMC Med Educ*. 2011;11:39.

**123**. Berg D, Berg K, Riesenberg LA, et al. The development of a validated checklist for thoracentesis. *Am J Med Qual*. 2013;28(3):220-226.

**124**. See KC, Jamil K, Chua AP, Phua J, Khoo KL, Lim TK. Effect of a pleural checklist on patient safety in the ultrasound era. *Respirology*. 2013;18(3):534-539.

**125**. Wayne DB, Barsuk JH, O'Leary KJ, et al. Mastery learning of thoracentesis skills by internal medicine residents using simulation technology and deliberate practice. *J Hosp Med*. 2008;3(1):48-54.

**126**. Bruce NC. Evaluation of procedural skills of internal medicine residents. *Acad Med*. 1989;64(4): 213-216.

**127**. Fox RA, Dacre JE, Clark CL, Scotland AD. Impact on medical students of incorporating GALS screen teaching into the medical school curriculum. *Ann Rheum Dis*. 2000;59(9):668-671.

**128**. Salamonsen M, McGrath D, Steiler G, Ware R, Colt H, Fielding D. A new instrument to assess physician skill at thoracic ultrasound, including pleural effusion markup. *Chest*. 2013;144(3):930-934.

**129**. American College of Emergency Physicians. AECP policy statement: emergency ultrasound guidelines. www.acep.org/NR/rdonlyres/ultrasound _guidelines.pdf. 2001. Accessed May 30, 2013.

**130**. American College of Surgeons. Ultrasound examinations by surgeons. www.facs.org/fellows _info/statements/st-31.html. 1998. Accessed May 30, 2013.

**131**. Feller-Kopman D. Ultrasound-guided thoracentesis. *Chest*. 2006;129(6):1709-1714.



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